Constructions — Notes

Watch the video and fill in each blank. Pause when you see “Your turn.”

Name: _____ Date: _____

☞ *Video: The two tools*

Key Idea

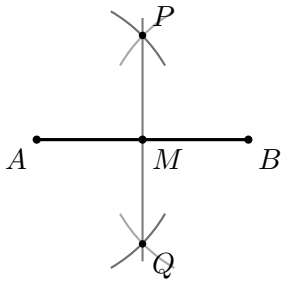
Compass: draws _____, and copies a _____.

Straightedge: draws straight lines — _____!

Four basics: copy a segment, copy an angle, bisect a _____, bisect an _____.

☞ *Video: Bisect a segment (find its midpoint)*

Key Idea



1. From A: open the compass _____ of AB. Draw arcs above and below.

2. From B: use the _____ width. Arcs above and below.

3. The arcs cross at ____ and ____.

4. Line PQ is the _____. It meets $\overline{AB}$ at the _____ M.

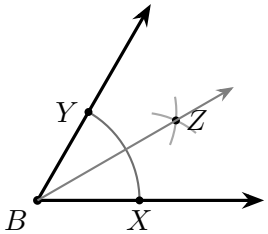
☞ *Video: Watch out*

Common Mistake

Keep the compass width the _____ for both arcs, and open it to _____ of AB — otherwise the arcs won’t _____.

☞ *Video: Bisect an angle*

Key Idea



1. An arc from the vertex B crosses both sides at ____ and ____.

2. Same width from X and from Y: the arcs meet at ____.

3. Ray BZ _____ the angle: two _____ halves.

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☞ *Video: Copy a segment*

Key Idea

Set the compass to _____. Put the point on C and draw an arc across the ray. Where it crosses is D.
Now CD _____ AB — copied without _____.

☞ *Video: Your turn*

Your Turn

You want to bisect a segment that is 8 cm long. Which compass width works: 3 cm or 5 cm?

Answer: _____, because it must be more than half of 8 cm, which is _____.
With 3 cm the arcs from A and B never _____.

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